



YouTube pinch-to-zoom

The new feature is currently available for Android smartphones with 18:9 aspect ratio
<http://dgit.in/YTFeat>

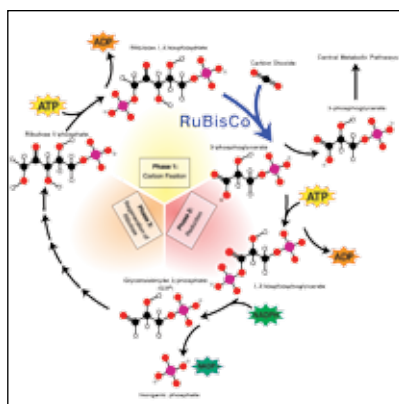


Jio's e-commerce move

Mukesh Ambani is reported going to use its Jio wireless service to hand out digital coupons
<http://dgit.in/JioShop>

how many paths there are. Plants are always growing new leaves and new cells and the system is very dynamic, so it seems obvious that there should be energy loss when transporting excitons, and yet there isn't. Turns out, the electron transport process in the reaction is almost perfect, which is more indicative of something quantum happening there, rather than a classical physics explanation.

This is where the probability wave that electrons use comes into play, and another effect we call quantum coherence. Basically, while classical physics thinks of it as the exciton traversing a



The Calvin cycle is pretty complicated...

path down from the chlorophyll molecules to the reaction center, quantum coherence suggests that the electron is a wave, and so is essentially finding the most efficient path, without losing energy. The excited electron is actually taking all the paths, and none of them, at the same time! (Don't blame us, everything quantum is weird.)

DO YOU SMELL THAT?

We've already overshot the word count for this article, but we cannot leave before mentioning something that isn't about birds and plants. There's quantum biology at work inside us as well, and one place it's being studied is inside your nose.

We're not going to go into depth here, so you might need to do some of your own research as well. The sense of smell that we have (or rather that all

animals capable of it have) works by having various receptors in the smelling apparatus (the nose in our case) which is like an incomplete jigsaw puzzle. Actually, it's like many different two piece jigsaw puzzles.

Imagine a few hundred molecules of something near you (dog poo, for instance) wafts into your nose. Your nose actually has receptors that have holes in the shapes of various commonly found chemical compounds and when a molecule fits into place, a signal is sent to your brain saying "found molecule xyz". Your brain has trained itself to recognise xyz as something (dog poo in our example), and then you go "eww"! Depending on the saturation of molecules entering your nose (which is a function of how close you are to something), the smell can be anything from faint to intense to even overpowering. This is the classical biology understanding of the sense of smell.

So why do some things with the exact same shape, still smell different then? This was what was found by researchers who replaced hydrogen in compounds with deuterium (a heavier hydrogen atom with a proton and a neutron). Chemically it's the same compound, has the same shape, and should fit perfectly into the same smell receptor in our noses, and yet it smells different. This has been confirmed with double blind tests, so it's apparent the shape theory of smell is sorely lacking. Quantum biology comes up with a hypothesis, and it has to do with the vibrations of the molecule. Because the mass of the deuterium molecule is higher, it vibrates differently. Could noses be sensing not just molecules but also how they vibrate? A theory suggests that receptors in noses might be utilising quantum tunnelling of electrons to detect vibrations of molecules that attach to them.

Imagine quantum tunnelling to be akin to wanting to throw a ball (an electron) into a house (another molecule) that has locked glass windows (barrier). Instead of breaking through the barrier and smashing the window, imagine if the ball just disappeared

from this side of the window and appeared inside the house. Because quantum mechanics suggests that electrons can be in two places at once, this is totally commonplace in the quantum world, and experiments have very much proven this. Tunnelling happens all the time when it comes to electrons or photons, and other quantum particles, so there's no reason to assume it is not happening inside our noses. What hasn't been proven is that it is the cause for us smelling heavy hydrogen elements differently.



Image Credit: TED.com

Biophysicist Luca Turin came up with the "vibrations" theory of smell

PROBLEMS

As you can see, the biggest problem with us being able to put these quantum biology theories to test is the problem of noise and complexity. We are just not skilled enough, and might never be capable of testing some of these theories, but we certainly should keep trying. We all know that the same quantum effects are responsible for a universe existing, when none should have, and there are quantum biologists who insist that the very reason for evolution and natural selection of life... mutations... might very well happen because of quantum effects.

For now, we put quantum biology under our ignorance section, but if all it proclaims happens actually does happen, we might have to put everything we think we know about biology into this section! Either way, there is so much to learn still. In science, ignorance is bliss, because it means big grant money!